

POCKETSMART AI: TECHNOLOGY STACK SPECIFICATION

1. Architectural Stack Layers

FRONTEND PRESENTATION: HTML5, CSS3, JavaScript (ES6+), Bootstrap 5.3	
TEMPLATE ENGINE: Jinja2 Server-Side Template Renderer	
BACKEND WEB FRAMEWORK: FastAPI 0.110+ (Python 3.12, ASGI Engine)	
APPLICATION SERVER: Uvicorn High-Performance ASGI Server	
AI / LLM LAYER: Google Generative AI Python SDK (Gemini Engine)	
AUTHENTICATION: Python-JOSE (JWT Tokens) & Passlib (Bcrypt Hashing)	
VALIDATION & SERIALIZATION: Pydantic v2 Schema Modeling	

DATA PERSISTENCE: Native Flat-File JSON Data Engine (users.json)
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2. Component Technology Evaluation & Justification

2.1 Backend: FastAPI & Python 3.12

- **Asynchronous Concurrency:** FastAPI leverages Python's asyncio primitives, handling concurrent client requests efficiently without blocking.
- **Automatic OpenAPI Documentation:** Native /docs and /redoc generation simplifies API testing and examiner evaluation.
- **Pydantic Validation:** Guarantees strict runtime data typing and error sanitization.

2.2 Artificial Intelligence: Google Gemini SDK

- **High Token Efficiency:** Rapid processing speeds ensure prompt responses for complex JSON extraction.
- **Multimodal Vision:** Natively interprets uploaded image streams (PIL.Image) alongside textual instructions within the same model call.

2.3 Presentation Layer: Jinja2 & Bootstrap 5

- **Zero Client Build Overhead:** Eliminates heavy Node.js/npm dependencies, enabling instant project startup directly through Python.
- **Mobile-First Responsive Grid:** Bootstrap 5 card layouts guarantee clean visual presentation across all device resolutions.

2.4 Security & Storage: OAuth2 JWT & users.json

- **Stateless Tokens:** JWT cookies permit secure endpoint protection without memory leaks.
- **Decoupled Persistence:** Flat-file JSON persistence ensures zero external database installations (e.g., MySQL or PostgreSQL) while preserving user state across server restarts.